Assignment 4: Sequence Modeling and Parsing

Academic Honesty: Please see the course syllabus for information about collaboration in this course. While you may discuss the assignment with other students, **your submission must be your own!**

Goals You will gain experience with basic structured prediction, including the Viterbi algorithm. You'll also look at parse trees, analyze syntactic phenomena, and reason about the behavior of PCFGs.

Deliverables and Submission Please type your solutions and do not submit handwritten work. You will submit your writeup to Gradescope as a PDF or text file of your answers to the questions.

Part 1: HMMs and Tagging (40 points)

Q1 (40 points) Consider the following corpus:

their/N raises/N
he/N raises/V
my/N purses/N

a) List the maximum-likelihood initial state probabilities for an HMM estimated from this data. (Hint: this should be a probability distribution over tags.)

b) List the maximum-likelihood transition probabilities for an HMM estimated from this data (including transitions to the STOP symbol).

c) Give an example of a grammatical English sentence **and its tags** using the words above that would be assigned zero probability under this HMM (assuming no smoothing is applied).

For the following question parts, assume that the log probabilities (log base 2) are as follows (these are rounded so they don't quite normalize as you would expect):

Initial	
N	-1
V	-1

Transitions	N	V	STOP
y_{i-1} = N	-1	-1.5	-2
y_{i-1} = V	-2	-2.5	-0.5

Emissions	their	he	my	raises	purses
N	-4	-2	-3	-3	-2
V	-5	-7	-5	-2	-4

Consider the sentence *he raises purses*

d) Apply the Viterbi algorithm on this data and show your work.

e) What is the highest posterior probability tag sequence for this sentence (now including the STOP transition)? **Is this sequence consistent with your interpretation of the sentence above?**

Part 2: Syntactic Parsers (60 points)

Q2 (20 points) In this part, you'll look at how parsers work in practice. You'll be using the demo for the Stanford CoreNLP constituency and dependency parsers—https://corenlp.run/—which are strong parsing models available as web demos. Under "Annotations" you should choose "part-of-speech" and either "constituency parse" and/or "dependency parse".

Dependency parsers produce relations between *head* words (parents) and *modifier* words (children). There are no intermediate categories like in constituency parsing; the sentence itself and its dependency arcs becomes a directed tree. Verbs are the heads of sentences. Subjects and objects of verbs are then modifiers of those verbs. Adjectives and nouns modify "head" nouns (e.g., for *art museum*, *museum* is the head and *art* is a modifier). These relationships are depicted as arrows going from parents to children.

a) For the sentences *I ate spaghetti with chopsticks* and *I ate spaghetti with meatballs*, describe the following. (1) Which of the two possible interpretations does the **constituency parser** choose for each sentence? Is it correct? (2) Do the same analysis for the **dependency parser**. (You should be able to understand the behavior by consulting the explanations of dependencies above.) (3) Are you surprised by the models' behavior here? Comment on what you see.

b) Find a new example that the **constituency parser** parses incorrectly. Include the example, its parse, and describe what is incorrect about the parse. Hint: you can think of what makes sentences ambiguous or try to find complicated sentences.

Q3 (20 points) Consider the following PCFG (bracketed numbers are probabilities):

NP $\rightarrow$ NP CC NP [0.3]
NP $\rightarrow$ NP PP [0.3]
NP $\rightarrow$ NNS [0.4]
PP $\rightarrow$ P NP [1.0]

NNS $\rightarrow$ cats [1.0]
CC $\rightarrow$ and [1.0]
P $\rightarrow$ in [1.0]

Define a PCFG with these rules, the nonterminals {NP, PP, NNS, CC, P}, terminals {*cats, and, in*}, and root symbol NP. For all question parts, provide justification so we can give partial credit as appropriate.

a) For the sentence *cats and cats*, how many valid syntactic parses (nonzero probability under this grammar) are there?

b) For the sentence *cats and cats in cats*, how many valid syntactic parses (nonzero probability under this grammar) are there?

Q4 (20 points) Consider the sentence *The cat chased the mouse through the garden*. Your task is to construct its dependency structure using the transition-based dependency parsing algorithm. You will make the classification decisions manually.

As you create links between words, you'll decide on the type of relationship (or dependency) they share. Please refer to the UD guidelines for possible relation types: `https://universaldependencies.org/u/dep/index.html`. Don't worry if you're unsure about the exact labels for each relation. Make your best guess based on the information available.

Show your work step by step.